Learning Theory and Knowledge Hierarchy for Artificial Intelligence Systems.

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***Abstract*—This work represents a continuation of our research in the realm of building Trustworthy AI and Explainable AI. The paper delves into the learning theory of intelligent systems, drawing upon classical mathematical approaches such as the task-based approach, the concept of semantic programming, computability theory, and model theory. This approach allows us to construct a knowledge base for AI systems, which can then be used to establish a natural hierarchy of knowledge within these systems. This enables AI systems to engage in continuous self-learning, as well as efficiently tackle the specific tasks they are designed for. Furthermore, AI systems can provide logical explanations for their decisions, accompanied by a set of guiding rules. This attribute of explainability, however, is notably absent in many large language models, leading us to explore the concept of hybrid AI systems, where neuro-symbolic AI integrates symbolic AI with neuro-statistical AI.**

***Index Terms*—theory of learning, hierarchy of knowledge, trustworthy artificial intelligence, neural networks, large language models, hybrid AI, LLM**

I. Introduction

The advent of artificial intelligence (AI) as a field of study can be traced back to the 1950s, when John McCarthy introduced the definition of the concept [1]. This novel direction piqued the interest of numerous scholars, and it quickly began to flourish. At this early stage, there were only rudimentary perceptrons [2], and there were no advanced language models based on transformer networks. Instead, classical methods from mathematics and programming dominated the field. It was predicted that within a decade or two, machines would effortlessly pass the Turing test [3], [4] and surpass human intellectual prowess. However, this time has passed, and machines have still not surpassed the intelligence of a 7-year-old child [5].

This situation can be attributed to several interrelated factors. Firstly, it concerned both the design of intelligent systems and their functionality. Moreover, there was a lack of a comprehensive learning theory specifically tailored for intelligent systems.

The logical and mathematical approaches were well-developed for addressing a narrow range of purely mathematical tasks. However, there was limited research into their application in solving problems that simulate human cognitive processes. Furthermore, the limited computational capabilities at the time made it challenging to validate mathematical methodologies and architectures that might prove effective.

A. Neural Networks and LLMs

The ubiquitous adoption of neural networks has been accompanied by the rapid advancement of computing technology. With the expansion of neural network architectures and the proliferation of training data, it has become evident that these systems exhibit enhanced intellectual capabilities and performance.

The process of training has evolved towards fine-tuning the parameters of neural networks through a variety of correction methods, including backpropagation. This methodology is also applicable to large language models built on transformer networks [6], which employ matrix representations rather than traditional neural network architectures.

Nonetheless, despite these advances, there remains a persistent challenge in enhancing the logical and interpretable aspects [7] of these neural network architectures [8]. While these systems can produce results and sometimes impress with their creativity, they do not necessarily strengthen these components in a meaningful way. Instead, they may only produce results that are similar to those produced by high school students, without truly understanding the underlying logic. However, when it comes to the clarity of their actions and the understanding of the meaning behind what is written, as well as solving simple mathematical problems, difficulties arise.

B. Brief description of the work

In the forthcoming sections, we will delve into the theory of developing intelligent systems, which is rooted in the task-based approach and the concept of semantic programming. This theory draws upon logical-probabilistic methods, empowering AI systems to effectively navigate problem-solving and generate meaningful outcomes.

The foundation of this theory lies in the concept of probabilistic knowledge, which will be defined below. A partial ordering is imposed on the set of probabilistic knowledge, allowing the system to identify the most optimal solution for

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a given task. Subsequently, we will explore the intricacies of generating novel knowledge and the challenge of uncovering the optimal solution. To evaluate the complexity, we will analyze the entire database, encompassing both expert knowledge and the knowledge acquired by the system during its operation.

II. Learning Theory

One of the primary metrics for assessing the quality of AI system is its learning capabilities and its capacity to effectively apply this knowledge in practical situations. Additionally, a crucial aspect is the system's ability to explain its decision-making process in a manner that is comprehensible to humans. This is particularly important in the context of developing trustworthy and explainable AI systems, as outlined by the four principles proposed by the National Institute of Standards and Technology (NIST) [9].

To date, there exists a plethora of systems for representing knowledge, including ontologies and knowledge graphs. In this particular section, we aim to introduce our own approach to knowledge representation, based on a family of first-order logical formulas. This approach allows us to treat knowledge as the set of objects that can be subjected to a partial ordering.

A. Preliminaries of Semantic Programming

Throughout this text, we shall assume that we operate within the constraints of the polynomial computable (p-computable) hereditary-finite list superstructure $HW(\mathfrak{M})$ of the signature σ [12] with list operations $head$, $tail$, $conc$, $cons$, $first$, $numElements$ and relationships $\in$, $\subseteq$, as well as a constant nil. Where the base elements of the base set of the $HW(M)$ structure encompass both the elements of the underlying in the base set of the p-computable structure M of the signature σ_0 and the hereditary-finite lists that are derived from these fundamental entities. Also, we assume that the standard model of arithmetic is embedded in $\mathfrak{M}$.

In this discourse, we shall primarily explore Δ_0-formulas in which the scope of quantifiers is restricted by relations $\in$, $\subseteq$, $\leq$. Furthermore, within the constraints of our linguistic framework, the notion of a term and formula will be inductively expanded through the use of conditional, p-iterative terms and modified p-iterative terms.

Within the constraints of the constructed language L, two significant propositions regarding polynomial complexity have been established: the polynomial analogue of Gandy's fixed point theorem [10], as well as a solution of the problem $P = L$ [11].

Within the concept of semantic programming, terms are essentially L-programs that receive a predetermined set of input data, and their execution occurs within the constraints of hereditary-finite lists superstructure $HW(\mathfrak{M})$ of the signature σ.

B. The probabilistic knowledge

The probabilistic knowledge is a triple:

$$K: \ (\forall x \exists y F(x,y), y = t(x), p) \tag{1}$$

where $\forall x \exists y F(x,y)$ is a problem (or task) which needs to be solved, $F(x,y)$ is a quantifying-free formula which has the form:

$$F(x,y) : \Phi(x,y) \to \Psi(x,y) \tag{2}$$

We say that $y = t(x)$ is solution of the problem $\forall x \exists y F(x,y)$ with probability p if for any interpretation γ where $\gamma(x) = a$, $a \in HW(M)$ the following is true:

$$Pr(HW(\mathfrak{M}) \models F(a, t(a)) \mid HW(\mathfrak{M}) \models \Phi(a, t(a))) = p$$

where Pr is a function of probability.

C. The conjunctive occurence

Let us consider two quantifier-free formulas $\Phi_1(\overline{x})$ and $\Phi_2(\overline{x})$ of the signature σ. We can define the binary relation

$$\Phi_1(\overline{x}) \subseteq_{\&} \Phi_2(\overline{x})$$

which establish the conjunctive occurrence of formula $\Phi_1(\overline{x})$ within formula $\Phi_2(\overline{x})$ by induction:

Base of induction: if Φ_1 does not have the form of $\Theta_1 \& \Theta_2$ for any formulas Θ_1 and Θ_2 then $\Phi_1 \subseteq_{\&} \Phi_2$ if and only if Φ_2 has the form $\Psi_1 \& \ldots \& \Phi_1 \& \ldots \& \Psi_n$ for some $\Psi_1, \ldots, \Psi_n$.

Step of induction let Φ_1 has the form $\Theta_1 \& \ldots \& \Theta_n \& \Theta_{n+1}$ then $\Phi_1 \subseteq_{\&} \Phi_2$ if and only if $\Theta_1 \& \ldots \& \Theta_n \subseteq_{\&} \Phi_2$ and $\Theta_{n+1} \subseteq_{\&} \Phi_2$.

Let us define the binary relation St is next:

$$St(\forall x \exists y F_1(x,y), \forall x \exists y F_2(x,y)) \Leftrightarrow$$

$$\Leftrightarrow \Phi_1(x,y) \subseteq_{\&} \Phi_2(x,y) \text{ and } \Psi_1(x,y) \subseteq_{\&} \Psi_2(x,y)$$

where $F_1(x,y)$ and $F_2(x,y)$ has the form (2).

Lemma 1: St is a polynomial computable relation.

Proof: To do this, it is enough to divide first formula into conjunctive members and check the occurrence of each member in the second formula.

All these operations are polynomial with respect to the sum of the lengths of these formulas. ■

D. The hierarchy of knowledge

Let us consider two probabilistic knowledge:

$$K_1: \ (\forall x \exists y F_1(x,y), y = t_1(x), p_1)$$

and

$$K_2: \ (\forall x \exists y F_2(x,y), y = t_2(x), p_2)$$

We define that K_2 is more powerful than K_1

$$K_1 \leq K_2$$

if and only if the following is true:

- $St(\forall x \exists y F_1(x,y), \forall x \exists y F_2(x,y))$
- $p_1 \leq p_2$

E. Facts and Knowledge creation

We call the fact d it is a triple:

$$d = (F(x,y), y = t(x), (a,b))$$

such that $b = t(a)$ and $HW(\mathfrak{M}) \models F(a,b)$, where $F(x,y)$ has the form 2.

F. Generation of probabilistic knowledge:

If we want to generate new knowledge

$$k = (\forall x \exists y (\Phi(x,y) \to \Psi(x,y)), y = t(x), p)$$

then we must check the fact base as follows:

1) We need to find all facts

$$d_i = (\Phi_i(x,y) \to \Psi_i(x,y), t(x), (a,b))$$

with the same solutions where $\Phi(x,y) \subseteq_{\&} \Phi_i(x,y)$ and $\Psi(x,y) \subseteq_{\&} \Psi_i(x,y)$. Let us denote the total number as n.

2) We need to find all facts

$$d_i = (\Phi_i(x,y) \to \Psi_i(x,y), t(x), (a,b))$$

with the same solutions where $\Phi(x,y) \subseteq_{\&} \Phi_i(x,y)$ and $\Psi(x,y) \not\subseteq_{\&} \Psi_i(x,y)$. Let us denote the total number as k.

Then the probability p will be found as the ratio $n/(n+k)$.

Otherwise, if there are too many facts, we can use different statistical criteria to find the probability.

G. Finding the best solution

Let us define a function f^* from two variables that finds the best solution $y = t(x)$ with the highest probability p from the knowledge base B for the problem $\forall x \exists y F(x,y)$. The algorithm itself that defines the function f^* can be described as follows:

1 new probabilistic knowledge K is consistently extracted from the knowledge base B.

2 next, the predicate $St(\forall x \exists y F(x,y), first(K))$ is checked

3 if the predicate St is true, then this knowledge is stored in a separate cell

4 if some other knowledge is found in the knowledge base on which the predicate $St(\forall x \exists y F(x,y), first(K'))$ will also be true, then the probability of this probabilistic knowledge is compared with the probability of the one stored in the cell.

5 if the probability p' is greater than the probability p, then the old knowledge is replaced by the new one

Theorem 1: Let $\forall x \exists y F(x,y)$ be the problem to be solved, B is the finite size knowledge base, then the function f^* of finding the most effective solution of the problem $\forall x \exists y F(x,y)$ within the knowledge base B will be a polynomial computable from the size of the knowledge base and the text length of the problem.

Proof: We must find the strongest solution (with highest probability) of the problem $\forall x \exists y F(x,y)$. To do so, we must take into account all probabilistic knowledge $K_i \in D_K$ in the form of

$$(\forall x \exists y F_i(x,y), y = t_i(x), p_i)$$

from the knowledge base that satisfies the p-computable relation $St(\forall x \exists y F(x,y), \forall x \exists y F_i(x,y))$, $i \in I$. Then we have to compare the effectiveness of these solutions among ourselves and choose the strongest one.

To prove that the function f^* is p-computable, it is sufficient to sequentially sort through the probabilistic knowledge in the knowledge base D_K that is consistent with problem $\forall x \exists y F(x,y)$, then this knowledge is compared with the most effective probabilistic knowledge that was found before.

This process is carried out iteratively, sequentially extracting new probabilistic knowledge from the knowledge base.

The computational complexity of the relation St is polynomial, which compares new probabilistic knowledge K with the most effective current probabilistic knowledge K_{ef} is polynomial by Lemma 1.

Suppose that $|\forall x \exists y F(x,y)| \leq |D_K|$. Since the sum of the lengths of all probabilistic knowledge K_i of the form $(\forall x \exists y F_i(x,y), y = t_i(x), p_i)$ from the knowledge base D_K does not exceed the size of the all knowledge base D_K we obtain the following inequality on complexity t:

$$\sum_{K_i \in D_K} t(St(\forall x \exists y F(x,y), \forall x \exists y F_i(x,y))) \leq$$

$$\leq \sum_{K_i \in D_K} C \cdot max(\forall x \exists y |F(x,y)|, \forall x \exists y |F_i(x,y)|)^p \leq$$

$$\leq |D_K| \cdot C \cdot |D_K|^p \leq C \cdot |D_K|^{p+1}$$

In another case if $|\forall x \exists y F(x,y)| \geq |D_K|$ the upper bound of the complexity is obtained:

$$C \cdot |\forall x \exists y F(x,y)|^{p+1}$$

In both cases, the overall computational complexity does not exceed:

$$C \cdot max(|\forall x \exists y F(x,y)|, |D_K|)^{p+1}$$

In case of comparing two probability knowledge from the knowledge base we have the following:

$$\sum_{K_i \in D_K} t(St(\forall x \exists y F_{j_i}(x,y)), \forall x \exists y F_i(x,y)) \leq$$

$$\leq C \cdot |D_K|^{p+1}$$

We can conclude that f^* is a p-computable function. ■

H. Logical rules of inference for knowledge

Let the following propositions $A \to B$ and $B \to C$ be true in first-order logic respectively, we can infer proposition $A \to C$ which will be also true. Would this proposition hold true if the same propositions were true with certain degrees of probability?

Probabilistic knowledge allows us to solve problems with a certain probability. Let $(\forall x \exists y \Phi(x, y) \to \Psi(x, y), y = t(x), p_1)$ and $(\forall x \exists y(\Psi(x, y)) \to \Theta(x, y)), y = t(x), p_2))$ are two probabilistic knowledge. Can we say something about the probability p_3 of next solution $y = t(x)$ for a problem: $\forall x \exists y(\Phi(x, y) \to \Theta(x, y))$? Given that p_1 and p_2 are both close to 1, is it reasonable to assume that p_3 will also approach 1?

The following lemma answer this question.

Lemma 2: For any p_1 and p_2 greater than 0 and less than 1 there exist two probabilistic knowledge $(\forall x \exists y(\Phi(x, y) \to \Psi(x, y)), y = t(x), p_1)$ and $(\forall x \exists y(\Psi(x, y)) \to \Theta(x, y)), y = t(x), p_2))$ such that for a new probabilistic knowledge $(\forall x \exists y(\Phi(x, y)) \to \Theta(x, y)), y = t(x), p_3))$ the value of p_3 can be anything in the range from 0 to 1.

Proof: This statement is proved by constructing two examples, when the probability of p_3 becomes close to 1 and when the probability of p_3 becomes close to zero.

Consider a polinomially computable finite model $\mathfrak{M}$ of the signature σ. Let the cardinality of the base set M be N.

Define the truth set of the formula on the model as follows:

$$A[\Phi(x, t(x))] = \{a \in M \mid \mathfrak{M} \models \varphi(a, t(a))\}$$

Define the truth set of the intersection of formulas on the model as follows:

$$A[\Phi(x, t(x)) \& \Psi(x, t(x))] = A[\Phi(x, t(x))] \cap A[\Psi(x, t(x))]$$

Then a new probabilistic knowledge

$$(\forall x \exists y(\Phi(x, y)) \to \Psi(x, y)), y = t(x), p))$$

have a value of probability p if and only if

$$\frac{A[\Phi(x, t(x)) \& \Psi(x, t(x))]}{A[\Phi(x, t(x))]} = p$$

Then, by changing the cardinality of the sets $A[\Phi(x, t(x))]$, $A[\Psi(x, t(x))]$, $A[\Theta(x, t(x))]$, $A[\Phi(x, t(x)) \& \Psi(x, t(x))]$ and $A[\Psi(x, t(x)) \& \Theta(x, t(x))]$ we can achieve probabilities p_1, p_2 close to 1 for probabilistic knowledge $((\forall x \exists y(\Phi(x, y)) \to \Psi(x, y)), y = t(x), p_1)$ and $((\forall x \exists y(\Psi(x, y)) \to \Theta(x, y)), y = t(x), p_2)$ respectively, but the probability p_3 for probabilistic knowledge $((\forall x \exists y(\Phi(x, y)) \to \Theta(x, y)), y = t(x), p_1)$ can be any from interval $(0, 1)$.

■

I. An open question

Now it is important to disscuss that in the case when we have probabilistic knowledge, many logic inference rules may not work correctly. Therefore, the question remains open, what restrictions can be found on probabilistic knowledge so that logical rules of inference can be set on their problems?

III. Conclusion

In this paper, we present the theory of training AI systems based on a logical-probabilistic approach, which involves continuously training AI systems in real-time using facts and expert data. We demonstrate how these systems apply their knowledge to solve problems that arise during their operation, ensuring that the solutions they provide are the most effective among those they are aware of.

These intelligent systems are easily integrated with large language models, enhancing their intellectual capabilities and enabling effective control over problem-solving processes. Moreover, they can be utilized to develop AI systems that adhere to specific rules within the framework of trustworthy artificial intelligence systems, making them suitable for applications in various domains such as digital replicas of smart cities, autonomous vehicle control, healthcare, finance, and other industries.

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